
Antonio Neto

📁 AI Engineer | Computer Science Student

🎓 Lattes

🐙 Github

✉ Email

🌐 LinkedIn

🌐 Website

👤 Professional Profile

I am a Brazilian AI Engineer and Computer Science student at PUC Minas, working with applied AI, optimization, infrastructure, and logistics problems. My research and development work covers LLM-based multi-agent systems, metaheuristics, and machine learning applied to industrial and logistics scenarios, with five accepted papers. I also study Mandarin at UFMG's Confucius Institute. I am highly focused, detail-oriented, and committed to everything I do.

🏢 Professional Experience

Deeptera **Apr 2026 – Present:** *Artificial Intelligence Engineer*
Part-time remote role focused on research and development

- Research and development on optimization problems and multi-agent systems.
- Designed architectures for intelligent systems integrated into conventional software.
- Authored papers accepted at SBPO and SIMPEP on optimization and multi-agent systems.

Deeptera **Aug 2024 – Apr 2026:** *Internship in Computing and Data*
Research and development

- Published article on multi-agent systems approved at SBAI.
- Participated in projects funded by FAPEMIG.
- Designed database architecture and Django backend using Azure and Azure DevOps infrastructure.
- Provided optimization and backend services for VLI.
- Conducted specialized development and research related to multi-agent systems, with application of RAG algorithms, personalized tools and scalable services.
- Applied genetic algorithms to solve optimization problems.

Hackatruck **2023:** *Project Participant*
Developed iOS applications as part of a month-long group project.

- Collaborated with team members to design and implement app functionalities.
- Used Swift, Node Red and Azure for development, handling various APIs.

📖 Publications

📄 **Uma Abordagem Baseada em Atenção para o Problema de Coletas e Entregas Dinâmicas.** A. Neto, R. D. Macedo, A. H. Duque, S. J. de Almeida. LVIII SBPO, 2026.
Attention Model trained with Reinforcement Learning and combined with VNS for the Dynamic Pickup and Delivery Problem, evaluated on the ICAPS 2021 benchmark.

📄 **DeepFactory: Sistema Multiagentes para Gestão e Otimização de Produção Industrial.** R. D. Macedo, A. Neto, A. H. Duque, S. J. de Almeida. LVIII SBPO, 2026.
Stage-oriented multi-agent platform integrating MIP scheduling, demand forecasting, predictive maintenance and computer-vision quality control.

 **Otimização do Sequenciamento de Operações de Carregamento e Descarregamento de Navios Graneleiros com Restrição de Trim via Algoritmo Genético.** A. Neto, R. D. Macedo, C. S. Almeida et al. (with VLI Multimodal). LVIII SBPO, 2026.

Genetic algorithm with trim constraints validated on 112 real operational plans from a multimodal logistics company.

 **Roteamento Multi-Período de Drones com Janelas de Tempo e Disponibilização Gradual de Estações de Recarga.** A. S. C. Neto, Z. K. G. do Patrocínio. SIMPEP, 2026.

Undergraduate thesis: MP-EAVNS, an energy-aware Variable Neighborhood Search for a multi-period extension of the E-VRPTW.

 **Proposta de um Sistema Multiagentes baseado em LLM: Integração em um Software de Gerenciamento e Otimização de Carregamento e Descarregamento de Navios.** A. Neto, C. S. Almeida, R. D. Macedo, A. H. Duque, S. J. de Almeida. SBAI/SBSE, 2025.

Hierarchical LLM multi-agent system (Manager, Helper, Optimizer and Data Analytics) integrated into a conventional ship loading and unloading system.

Education

PUC Minas

2023 – Present: *Bachelor of Computer Science*

I have worked with various technologies at PUC and specialized in machine learning, data science, and AI technologies outside of college. My undergraduate thesis addresses multi-period drone routing with time windows and gradual charging-station deployment, advised by Prof. Zenilton K. G. do Patrocínio.

Confucius Institute (UFMG)

2024 – Present: *Mandarin Student (HSK 4)*

I study Mandarin out of a deep appreciation for the culture and its relevance to the field of AI.

Key Technical Skills

- ✓ **Programming Languages:** Python, C, TypeScript
- ✓ **Agent Systems:** LangChain, LangGraph, RAG, MCP
- ✓ **Machine Learning:** PyTorch, Scikit-learn, XGBoost
- ✓ **Data Science:** Pandas, NumPy, Matplotlib, Seaborn
- ✓ **Backend:** Django, FastAPI, REST APIs
- ✓ **Databases:** PostgreSQL, MySQL
- ✓ **DevOps:** Azure DevOps, Docker
- ✓ **Version Control:** Git, GitHub
- ✓ **Optimization:** AMPL, pyomo, GLPK/SCIP solvers, metaheuristics (GA, VNS, ALNS, GRASP)
- ✓ **Cloud:** Azure, Google Cloud

Certifications

- ⚙️ Cloud Services Practices using Swift with emphasis on Cognitive Services
- ⚙️ Algorithms and OOP with Swift, JavaScript, and RESTful APIs
- ⚙️ Agriculture 4.0: Optimizing with Artificial Intelligence

Languages

- 🌐 Portuguese - Native
- 🌐 English - Fluent
- 🌐 Mandarin - HSK 4